

PA5: Dermatological Image Classification

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1 Introduction

In this assignment, we are given images of skin marks that are classified into 2 types: melanoma and melanocytic nevus. We will be looking into 5 different models that will perform image classification on the provided images. We will compare these model with each other on the validation dataset and finally compare the best model on the test set. We will also look at some limitation and ethical considerations regarding this assignment.

2 Methodology

For this assignment, we will be training a baseline model and test it on the validation set. We will be creating 4 other models and compare their accuracy against each other and the baseline models, the 4 other models are: baseline model with normalisation, baseline model with residual connections, baseline model with data augmentation and a transfer learning model (VGG16 model). After testing all the models on the validation set and comparing their accuracies, we will use the best model and run it on the test set.

3 Results

In this section, we will look at the results of the models after running them on the validation set. In the last subsection, we will look at the best model and what accuracy it achieved when it was run on the test set.

3.1 Baseline Model

We obtained an accuracy of 80.59% on the baseline model that we made.

Figure 1 shows the Confusion Matrix of the Baseline Model obtained when it was run against the validation set.

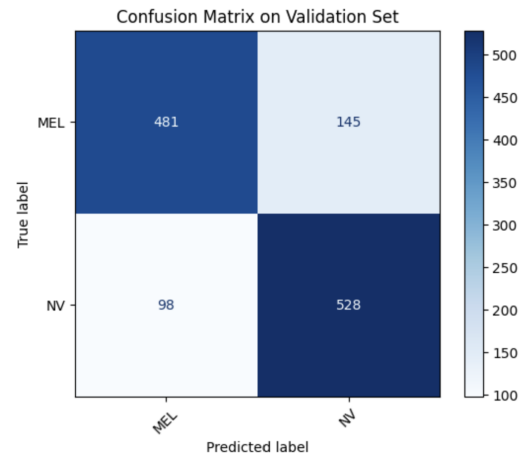


Figure 1: Baseline Model Validation Confusion Matrix

3.2 Baseline Model with Normalisation

We obtained an accuracy of 80.59% on the baseline model with normalization.

Figure 2 shows the Confusion Matrix of the Baseline with Normalisation Model obtained when it was run against the validation set.

3.3 Baseline Model with Residual Connections

We obtained an accuracy of 79.79% on the baseline model with residual connections.

Figure 3 shows the Confusion Matrix of the Baseline with Residual Connections Model obtained when it was run against the validation set.

3.4 Baseline Model with Data Augmentation

We obtained an accuracy of 76.44% on the baseline model trained with augmented data.

Figure 4 shows the Confusion Matrix of the Baseline with Data Augmentation Model obtained when it was run against the validation set.

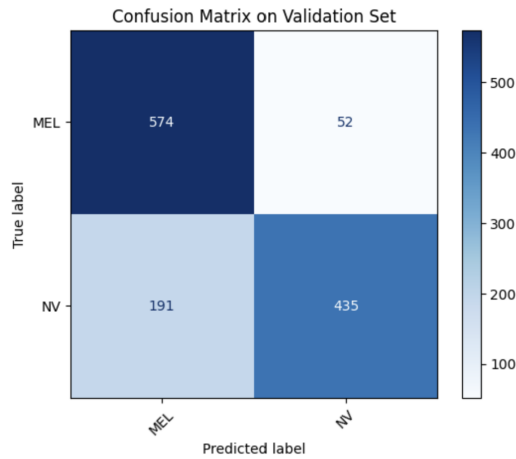


Figure 2: Baseline with Normalisation Model Validation Confusion Matrix

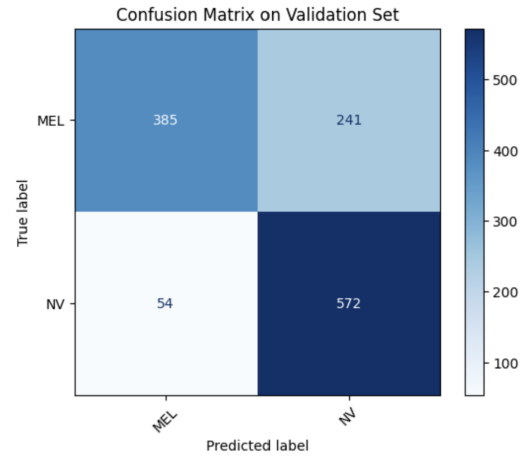


Figure 4: Baseline with Data Augmentation Model Validation Confusion Matrix

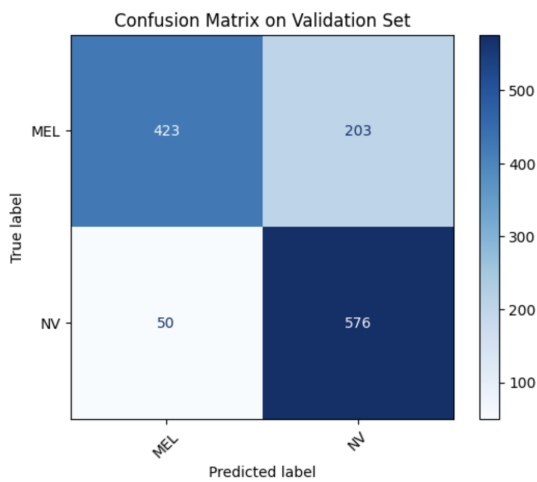


Figure 3: Baseline with Residual Connections Model Validation Confusion Matrix

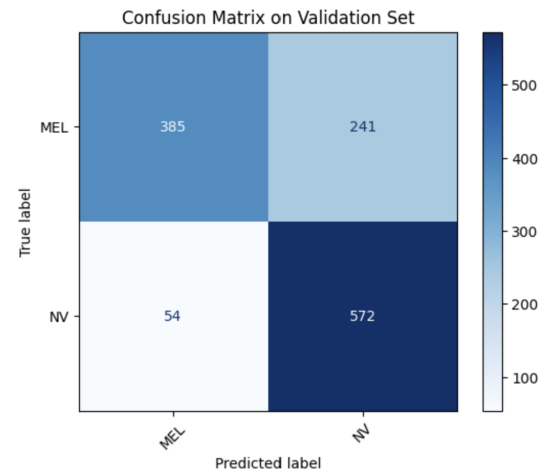


Figure 5: VGG16 Model Validation Confusion Matrix

3.5 Transfer Learning Model

We obtained an accuracy of 76.44% on the pre-trained model (VGG16 model).

Figure 5 shows the Confusion Matrix of the Transfer Learning (VGG16) Model obtained when it was run against the validation set.

3.6 Baseline with Normalisation on Test Set

Based on the results obtained from comparing the models on the validation set, we found that the best model we had was the Baseline Model and the Baseline Model with Normalisation. Given that Batch Normalisation can help models be more robust to noise, we decided to choose the Baseline Model with Normalisation over just the Baseline Model in the case whereby the validation accuracy

of only using the Baseline Model is a fluke. We will be running this model against the test set. We obtained an accuracy of 81.11% when we ran it on the test set.

Figure 6 shows the Confusion Matrix of the Baseline with Normalisation Model obtained when it was run against the test set.

4 Conclusion

In this assignment, we successfully built and compared our own models that could classify and distinguish images of melanoma and melanocytic nevus with the best model obtaining an accuracy of 81.11% when it ran against the test set. For future continuation of this assignment, more complex models which specialize in computer vision could be used. This could possibly generate bet-

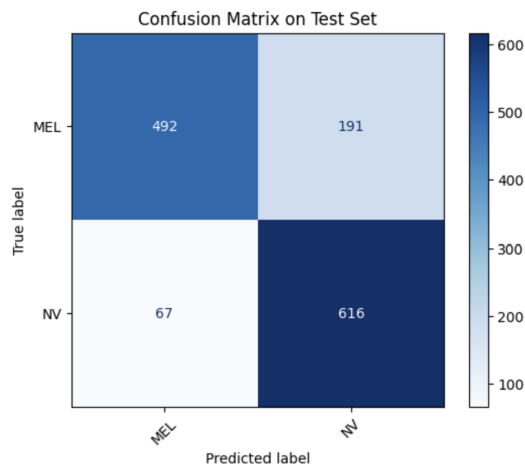


Figure 6: Baseline with Normalisation Model Test Confusion Matrix

ter results than what was obtained in this assignment. Furthermore, more hyperparameter tuning could have been done to further improve the results.

5 Limitations

Some limitations that we encountered in this assignment are resources for training the models. Our training time for the models took a long time as the models were being trained on images. Another limitation would be that the models are not close to 100% accuracy, which prevents doctors to fully rely on these models in a real world situation.

6 Ethical Considerations

An ethical consideration we can think of would be the usage of these models. Since these models are not close to 100% accuracy, doctors cannot fully rely on such models to help them with the diagnosis of melanoma or melanocytic nevus. If they were to fully rely on such models, there would be a chance of misdiagnosis, and this could affect the patient greatly.